

II. Chemistry

Atomic Structure

Notes

Kanad (कणाद) :

- Kanad, also known as Kashyap, Uluka and Kanabhuk, was an Indian natural scientist and Philosopher, who founded the **Vaishesika School of Indian Philosophy**.
- Estimated to have lived sometimes between 6th century to 2nd century BC, little is known about his life. He is known for developing the foundation of an atomistic approach of the universe.
- He suggested that everything can be subdivided, but this subdivision can not go on forever and there must be smallest entities (Parmanu) that cannot be divided, that are eternal, that aggregate in different ways to yield complex substances and bodies with unique identity, a process that involves heat and this is the basis for all material existence.

Dalton's Atomic Theory :

- In the beginning of 19th century, Dalton proposed that all matters are composed of atoms which are indivisible and indestructible building blocks.
- All atoms of an element are identical.
- Different elements have atoms of different size and mass.
- Atoms are neither created nor could be destroyed. This implies that during chemical reactions, no atoms are created or destroyed.
- Atoms of different elements combine in a fixed ratio to form compounds.
- The formation of new products (compounds) results from the rearrangements of existing atoms.

Drawbacks of Dalton's Atomic Theory –

- It was proved that an atom is not indivisible. An atom can be subdivided into electrons, protons and neutrons. But remember that atom is the tiniest particle that takes part in a chemical reaction.

Fundamental Particles of Atom :

(i) Electron-

- An electron is negatively charged sub-atomic particle.
- It can be either free (not attached to any atom) or bound to the nucleus of an atom.

- Electrons in atoms exist in spherical shells of various radii, representing energy levels.
- The charge on an electron is (-) 1.6022×10^{-19} coulomb.
- Its symbol is e^- or β^- .
- The electron has a mass that is approximately $1/1837$ that of the proton or 9.109×10^{-31} kg.
- It was discovered by J.J. Thomson (1897).
- Atomic mass unit of the electron is 0.0005486.

(ii) Proton –

- A proton is a positively charged sub-atomic particle.
- Its symbol is p or p^+ .
- The charge on a proton is (+) 1.6022×10^{-19} coulomb.
- Its weight is 1.6726×10^{-27} kg.
- Its atomic mass unit is 1.00727.
- It was first discovered by Goldstein as anode rays (1886) and finally identified and named by Rutherford (1919).

Note : The number of electrons is equal to the number of protons in an atom with an equal charge of opposite nature. Hence the atom is neutral.

(iii) Neutron –

- It is a neutral sub-atomic particle so its name is neutron.
- It is represented by symbol n or n^0 .
- Its mass is slightly larger than that of proton i.e 1.0087 amu.
- It was discovered by James Chadwick (1932).

Note : Atomic mass unit (amu) is defined as precisely $1/12$ of the mass of an atom of Carbon-12.

Constitution of Atomic Nucleus :

- The atomic nucleus is the small, dense region consisting of proton and neutrons at the center of an atom.
- The proton & neutron jointly called nucleon.
- It was discovered by Ernest Rutherford (1911) based on the Geiger - Marsden gold foil experiment (1909).
- Protons and neutrons are bound together to form a nucleus by the nuclear force.

Atomic Number (Z) :

- The symbol of atomic number (Z) is a conventional symbol possibly came from the German word Atomzahl which means atomic number.
- Atomic number of an element is the number of protons in the nucleus of its atom.
- It is the characteristic of a chemical element and determines its place in the periodic table.
- The atomic number of each element is certain and constant.
- Atoms of different elements are with different atomic number.
- The atomic number of Hydrogen is 1, it means it has one proton in its nucleus.

Mass Number (A) :

- (A) has been derived from German word Atomge which means atomic weight.
- The total number of protons and neutrons in an atom's nucleus is known as its mass number.
- It determines the atomic mass of atoms.
- A mass number of atom = No. of protons + No. of neutrons in nucleus of an atom.
- It represents the total number of nucleons in the nucleus of an atom.
- The mass number is not a fundamental character of the element.
- An element may have atoms with different mass number.

Atomic Weight :

- The atomic weight of an atom is approximately equal to its mass number.

$$\text{Atomic weight} \approx \text{Mass number (A)}$$

$$\text{Atomic weight} \approx \text{No. of protons in nucleus} + \text{No. of neutrons in the nucleus.}$$

Isoelectronic :

- The ions and atoms with an equal number of electrons are known as isoelectronic.
- The electronic configuration of these ions and atoms are the same. For example –
 - a. $\text{Ne} \rightarrow 2,8$
 - b. $\text{Na}^+ \rightarrow 2,8$
 - c. $\text{Mg}^{++} \rightarrow 2,8$
 - d. $\text{F}^- \rightarrow 2,8$
 (Each ion/atom with same electronic configuration of 10 electrons)

Isotopes :

- Isotopes are atoms of a chemical element that have the same atomic number but are with different atomic mass.
- The place of isotopes is same in the periodic table as they have the same atomic number.

Isotopes of Hydrogen (three isotopes) –

Isotope	No. of protons/ Atomic Number	No. of neutrons	Atomic Mass
Protium (${}_1\text{H}^1$)	1	0	1
Deuterium (${}_1\text{H}^2$)	1	1	2
Tritium (${}_1\text{H}^3$)	1	2	3

Isotopes of oxygen – There are three isotopes of oxygen viz. ${}_8\text{O}^{16}$, ${}_8\text{O}^{17}$, ${}_8\text{O}^{18}$.

Isobars :

- Isobars are atoms of different chemical elements having the same number of nucleons (Atomic mass), but their atomic numbers are different.
Example – ${}_1\text{H}^3$ & ${}_2\text{He}^3$
- The place of isobars is different in the periodic table because their atomic number differs.

Isoneutronic or Isotones :

- Different element's atom with the same number of neutrons are called Isoneutronic or Isotones.
Example –
 - a. ${}_1\text{H}^3$ and ${}_2\text{He}^4$ – Both have same number of neutrons i.e. 2 neutrons.
 - b. ${}_{14}\text{Si}^{30}$, ${}_{15}\text{P}^{31}$, ${}_{16}\text{S}^{32}$ – All with the same number of neutrons i.e. 16 neutrons.
 - c. ${}_6\text{C}^{13}$, ${}_7\text{N}^{14}$ – both have the same number of neutrons i.e. 7 neutrons.

Radioactivity :

- Radioactivity is a natural phenomenon which occurs in unstable elements.
- These elements have spontaneous disintegration character.
- Such elements are called radioactive substances.
- α , β & γ rays are emitted from the atoms of radioactive substances. These are known as radioactive rays.
- Radioactivity is a nuclear reaction, not a chemical reaction.
- It was discovered by Henry Becquerel in Uranium.
- In 1898 Marie Curie and her husband Pierre Curie discovered a radioactive element Polonium.
- In 1902, Marie Curie discovered Radium from the ore of uranium Pitch Blend (U_3O_8).

Difference between alpha, beta & gamma rays

S.N.	Character	α -Rays	β -Rays	γ -Rays
1.	Nature	(i) These are nuclei of helium. (ii) These are with two units of positive charge. (iii) Its mass is equal to the atomic mass of nucleus of helium atom (4 amu). (iv) It is represented by ${}_2\text{He}^{4++}$, α^{++} or α .	(i) These are fast-moving electrons. (ii) They carry negative charge. (iii) Its weight is 0.0005486 amu. (iv) It is represented by $-\beta^0$ or ${}_1e^0$.	(i) These are electromagnetic radiations. (ii) These are neutral in nature.
2.	Velocity	Velocity of alpha (α) rays is approximately equal to 1/10th of the velocity of light.	Velocity of beta (β) rays is approximately equal to the velocity of light.	Velocity of gamma (γ) rays is equal to the velocity of light.
3.	Penetrating Power	Penetrating power is very low. These can penetrate 0.002 cm thick Aluminium sheet.	Penetrating power is much greater than alpha rays due to their high speed. These can penetrate 0.2 cm. thick Aluminium sheet.	Due to non - material nature (Electromagnetic radiation) and very high speed the penetrating power of these rays are greater. These rays can penetrate 100 cm thick Aluminium sheet.
4.	Ionising Power	α -rays have very high power to ionize gases.	β -rays ionise gases to lesser extent. 100 times less than α -rays.	These are weak ionizer of gases. 100 times less than β -rays.
5.	Effect of electric and magnetic field	α -rays are deflected towards negative charged plate. α -rays are also deflected in magnetic field.	β -rays are deflected towards positive charged plate. β -rays are also deflected in magnetic field.	γ -rays are not deflected in electric and magnetic fields.

Question Bank

1. Who is regarded as the Father of Modern Chemistry?

- Rutherford
- Einstein
- Lavoisier
- C.V. Raman
- None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (c)

French nobleman and chemist Antoine Lavoisier is regarded as the 'Father of Modern Chemistry'. Lavoisier is most noted for his discovery of the role oxygen plays in combustion. He recognized and named oxygen (1778) and hydrogen (1783), wrote the first extensive list of elements and helped to reform chemical nomenclature.

2. Dalton's atomic theory successfully explained :

- Law of conservation of mass.
- Law of constant composition.
- Law of radioactivity.
- Law of multiple proportion.

- (i), (iii) and (iv)
- (i), (ii) and (iii)
- (i), (ii) and (iv)
- (i), (iii) and (iv)

70th B.P.S.C. (Pre) 2024

Ans. (c)

John Dalton's atomic theory states that all matter is made of atoms, which are indivisible particles. Dalton's theory was based on experiments and observations of gases. Postulates of Dalton's atomic theory are as follows :

- All matter is made of atoms.
- Atoms of the same element are identical in size, mass, and other properties.

- Atoms of different elements differ in size, mass, and other properties.
- Atoms cannot be created or destroyed.
- Compounds are formed when two or more different types of atoms combine.
- A chemical reaction is a rearrangement of atoms.

Dalton's theory provided a scientific basis for understanding why elements combine in certain proportions to form compounds. It also explained the law of constant composition, conservation of mass, and the law of multiple proportions.

3. Who was the first one to propose a model for the structure of an atom ?

- (a) E. Goldstein (b) Rutherford
(c) Neils Bohr (d) J.J. Thomson

70th B.P.S.C. (Pre) 2024

Ans. (d)

J.J. Thomson was the first scientist to propose a model for the structure of an atom, suggesting a positively charged sphere with negatively charged electrons embedded within it, often referred to as the 'plum pudding' model. Thomson's model, proposed in the late 19th century, envisioned the atom as a sphere of positive charge with negatively charged electrons scattered throughout, similar to plums in a pudding. Thomson's model was a direct result of his discovery of the electron, a subatomic particle with a negative charge, which challenged the previous understanding of the atom as a fundamental, indivisible unit. Thomson's model was later superseded by more sophisticated models, such as Rutherford's nuclear model (with a positively charged nucleus and orbiting electrons), and then Bohr's model, which introduced the concept of quantized energy levels.

4. The positively charged part at the centre of an atom is called as :

- (a) Proton (b) Neutron
(c) Electron (d) Nucleus
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (d)

The positively charged part at the centre of an atom is called as nucleus. The atomic nucleus is the small, dense region consisting of proton and neutron at the centre of an atom.

5. Constituents of atomic nucleus are –

- (a) Electron and proton
(b) Electron and neutron

- (c) Proton and neutron
(d) Proton, neutron and electron

U.P. P.C.S. (Pre) 1996

41st B.P.S.C. (Pre) 1996

43rd B.P.S.C. (Pre) 1999

Ans. (c)

The constituents of an atom are protons, neutrons and electrons. The protons and neutrons (nucleons) are found in the nucleus of atoms. The nucleus of an atom is surrounded by electrons.

6. The nucleus of an atom consists of :

- (a) electrons and neutrons (b) electrons and protons
(c) protons and neutrons (d) electrons only
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (c)

See the explanation of above question.

7. Which among the following is an incorrect option?

- (a) Atom is electrically neutral due to presence of protons and electrons inside the nucleus.
(b) Electrons are negatively charged and have negligible mass.
(c) Atomic number is the number of protons in an atom of an element.
(d) Isotopes are useful as nuclear fuel, in medical field, in carbon dating, etc.
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (a)

An atom consists of a positively charged nucleus, surrounded by one or more negatively charged particles called electrons. The atomic nucleus consists of protons and neutrons while electrons revolve around the nucleus. The positive charges in an atom equal the negative charges, so the atom has no overall charge, it is electrically neutral. Statements of options (b), (c) and (d) are correct.

8. Which of the following statement about molecular structure is correct :

- (a) Neutron and electron are found inside the nucleus and protons revolves around the nucleus.
(b) Electron and proton are inside the nucleus and neutron revolve around the nucleus.

- (c) Proton and neutron are inside the nucleus and electrons revolve around the nucleus.
(d) Proton, neutron and electron all are inside the nucleus.

R.A.S./R.T.S. (Pre) 1992

Ans. (c)

See the explanation of above question.

9. Which of the following is not a part of an atom?

- (a) Electron (b) Proton
(c) Neutron (d) Photon

M.P.P.C.S. (Pre) 2005

Ans. (d)

Proton, electron and neutron are part of an atom, but the photon is associated with light energy and also known as energy packet of light.

10. A single type of atom is found in –

- (a) Compounds of minerals
(b) Mixture of minerals
(c) Native elements
(d) None of the above

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

Native element is a material that consists of a single type of atom, while a compound consists of two or more types of atoms.

11. The atomic nucleus was discovered by :

- (a) Rutherford (b) Dalton
(c) Einstein (d) Thompson

R.A.S./R.T.S.(Pre) 2003

Ans. (a)

The nucleus is a very dense region consisting of protons and neutrons at the centre of an atom. It was discovered in 1911 as a result of Ernest Rutherford's interpretation of the 1909 Geiger–Marsden gold foil experiment. The proton–neutron model of the nucleus was proposed by Dmitri Ivanenko in 1932.

12. Consider the following events and arrange them in correct chronological order starting from the earliest to the last activity :

- (I) Discovery of pi meson
(II) Discovery of neutron
(III) Discovery of electron
(IV) Discovery of proton

Select the correct answer from the code given below :

Code :

- (a) (III), (II), (I), (IV) (b) (III), (IV), (II), (I)

- (c) (I), (III), (IV), (II) (d) (III), (IV), (I), (II)

U.P. R.O./A.R.O. (Pre) 2023

Ans. (*)

- (I) Discovery of pi meson – 1947 (by Cecil Powell)
(II) Discovery of neutron – 1932 (by James Chadwick)
(III) Discovery of electron – 1897 (by J.J. Thomson)
(IV) Discovery of proton – 1886 (by Goldstein)

Hence, the correct chronological order is IV, III, II, I which is not given in any option.

Note : Proton was first discovered by Goldstein as anode rays (1886) and finally identified and named by Rutherford (1919). If the credit of proton discovery is given to Rutherford then the correct chronological order will be III, IV, II, I which is given in option (b).

13. In atom, neutron was discovered by :

- (a) J.J. Thomson (b) Chadwick
(c) Rutherford (d) Newton

Jharkhand P.C.S. (Pre) 2003

U.P.P.C.S. (Pre) 1996

U.P.P.C.S. (Pre) 1995

Ans. (b)

Neutron is a neutral particle which has no electrical charge. It was discovered by Chadwick in 1932. He discovered that when Beryllium (Be) or other atoms are bombarded with high speed particles, then electrically neutral particles emerges from these atoms and such neutral particles are known as neutrons.

14. Which of the following is not a sub-atomic particle?

- (a) Neutron (b) Proton
(c) Deuteron (d) Electron

U.P.P.C.S. (Pre) 2007

Ans. (c)

The particles that are smaller than the atoms are called sub-atomic particles. The three main sub-atomic particles that form an atom are protons, neutrons and electrons. Deuteron contains a proton and a neutron.

15. Which one of the following pairs constitutes particle-antiparticle pair?

- (a) Electron - Positron (b) Proton - Neutron
(c) Photon - Electron (d) Neutron - Neutrino

U.P.P.C.S. (Mains) 2005

Ans. (a)

The positron or antielectron is the antiparticle or the antimatter counterpart of the electron. The positron has an positive electric charge and has the same mass as an electron.

16. Match List-I with List-II and select the correct answer using the codes given below the lists :

List-I (Characteristic)	List-II (Particle)
A. Zero mass	1. Positron
B. Fractional charge	2. Neutrino
C. Fractional spin	3. Quark
D. Integral spin	4. Phonon

Code :

	A	B	C	D
(a)	2	3	1	4
(b)	3	2	4	1
(c)	2	3	4	1
(d)	3	2	1	4

I.A.S. (Pre) 2001

Ans. (a)

The positron has a positive electric charge (+1e) and has the same mass and a spin of 1/2 as an electron. Neutrinos are sub-atomic particles produced by the decay of radioactive elements and are elementary particles that lack an electric charge and a very small mass, which might even thought to be zero. Quarks combine to form composite particles called hadrons, the most stable of which are protons and neutrons and are the components of atomic nuclei. They have fractional charge of +2/3e or -1/3e. Phonon, in condensed-matter physics, a unit of vibrational energy that arises from oscillating atoms within a crystal. A phonon is a definite discrete unit or quantum of vibrational mechanical energy, just as a photon is a quantum of electromagnetic or light energy. Phonon has integral spin like photon.

17. Which one is a fundamental particle?

- | | |
|------------------|------------|
| (a) Neutron | (b) Proton |
| (c) π -meson | (d) Quark |

U.P. R.O./A.R.O. (Mains) 2017

Ans. (d)

In particle physics, an elementary particle or fundamental particle is a subatomic particle with no sub-structure, thus not composed of other particles. Particles currently thought to be elementary include the fundamental fermions (quarks, leptons, antiquarks and antileptons) as well as fundamental bosons.

18. Higgs Boson are also known as :

- | | |
|--------------|-------------------|
| (a) Electron | (b) Proton |
| (c) Neutrons | (d) God Particles |

Jharkhand P.C.S. (Pre) 2021

Ans. (d)

The Higgs Boson is the fundamental force-carrying particle associated with the Higgs field, a field that gives mass to other fundamental particles such as electrons and quarks. In the mainstream media, the Higgs boson has often been called the 'God Particle' from the 1993 book 'The God Particle : If the Universe is the Answer, What is the Question?' by Nobel Laureate Leon M. Lederman and science writer Dick Teresi.

19. Much discussed 'God Particle' is :

- | | |
|-----------------|-------------|
| (a) Neutrino | (b) Leptons |
| (c) Higgs boson | (d) Mesons |

U.P. U.D.A./L.D.A. (Pre) 2013

Ans. (c)

The Higgs Boson is an elementary particle in the standard model of particle physics. The hypothesis of its existence was given in 1964 but practically it was proved on March 14, 2013. In mainstream media the Higgs Boson has often been called the 'God particle'. The two physicists who discovered these particles Peter Higgs and Francois Englert were awarded the Nobel Prize in Physics in 2013.

20. 'God particle' is :

- | | |
|--|-----------------|
| (a) Neutrino | (b) Higgs Boson |
| (c) Meson | (d) Positron |
| (e) None of the above/More than one of the above | |

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (b)

See the explanation of above question.

21. The efforts to detect the existence of Higgs boson particle have become frequent news in the recent past. What is/are the importance of discovering this particle?

1. It will help us to understand as to why elementary particles have mass.
2. It will enable us in the near future to develop the technology to transfer matter from one point to another without traversing the physical space between them.
3. It will enable us to create better fuel for nuclear fission.

Select the correct answer using the codes given below:

- | | |
|------------------|---------------------|
| (a) only 1 | (b) only 2 and 3 |
| (c) only 1 and 3 | (d) only 1, 2 and 3 |

I.A.S. (Pre) 2013

Ans. (a)

The Higgs boson is the fundamental force-carrying particle of the Higgs field, which is responsible for granting other particles their mass. The study of this field is important to know that why some fundamental particles have mass. So the discovery of Higgs boson particles is important to understand this fact.

22. Which particle is free of charge?

- (a) α -particle
- (b) Electron
- (c) Neutron
- (d) Proton
- (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (c)

Among the given particles neutron is free of charge. α (alpha) particle (+2ve) and proton (+ve) are positive charged particles while electron (-ve) carries negative charge.

23. Which of the following particles has zero charge?

- (a) Positron
- (b) Neutrino
- (c) Electron
- (d) α -particle

U.P. P.C.S. (Pre) 2021

Ans. (b)

A neutrino is a subatomic particle that is very similar to an electron, but has no electrical charge and a very small mass, which might even be zero. Neutrinos are one of the most abundant particles in the universe. Because they have very little interaction with matter, however, they are incredibly difficult to detect.

24. The alpha particle carries two positive charges. Its mass is very nearly equal to that of –

- (a) Two protons
- (b) Nucleus of an atom of helium
- (c) Sum of masses of two positrons and two neutrons
- (d) Two positrons, as each positron carries a single positive charge

I.A.S. (Pre) 1996

Ans. (b)

An alpha particle is a fast moving particle containing two protons and two neutrons (a helium nucleus). Its mass is equal to that of helium atom's nucleus. Alpha particles carry a charge of +2 and strongly interact with matter.

25. The Nucleus of Helium has –

- (a) Only one neutron
- (b) Two protons
- (c) Two protons and two neutrons
- (d) One proton and two neutrons

R.A.S./R.T.S. (Pre) 1996

Ans. (c)

See the explanation of above question.

26. Which of the following elements does not contain neutrons?

- (a) Oxygen
- (b) Nitrogen
- (c) Hydrogen
- (e) Copper
- (d) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (c)

The atomic no. of Hydrogen is 1. The Protium (${}_1\text{H}^1$) isotope of the Hydrogen has no neutron while Deuterium (${}_1\text{H}^2$) has one neutron and Tritium (${}_1\text{H}^3$) isotope has 2 neutrons.

27. Which of the following carries a negative charge?

- (a) X-rays
- (b) Alpha particles
- (c) Beta particles
- (d) Gamma rays

45th B.P.S.C. (Pre) 2002

Ans. (c)

Alpha rays or Alpha particles (α) are the positively charged particles. Beta particles (β) are highly energetic electrons which are released from inner part of a nucleus. They are negatively charged (-1e) and have a negligible mass. Gamma radiation (γ) consist of photons, which travel at the speed of light like all electromagnetic radiations. A ray has no mass or charge. Gamma radiation can travel much faster (speed of light) in the air than alpha and beta.

28. Which among the following is a positively charged particle emitted by a radioactive element?

- (a) Cathode ray
- (b) Beta ray
- (c) Alpha ray
- (d) More than one of the above
- (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

Alpha rays or Alpha particles (α) are the positively charged particles emitted by a radioactive element. These are nuclei of helium (${}_2\text{He}^{4++}$). Its mass is equal to the atomic mass of nucleus of helium atom.

29. In an atom, the order of filling up of the orbitals is governed by –

- (a) Aufbau's principle
- (b) Heisenberg's uncertainty principle
- (c) Hund's rule
- (d) Pauli's exclusion principle

I.A.S. (Pre) 2001

Ans. (a)

The Aufbau principle is used to determine the electron configuration of an atom, molecule or ion. The principle postulates a hypothetical process in which an atom is 'built up' by progressively adding electrons. As they are added, they assume their most stable condition (electron orbitals) with respect to the nucleus and those electrons already there. According to the principle, electrons fill orbitals starting at the lowest available (possible) energy levels before filling higher levels (e.g. 1s before 2s).

30. Regarding the atom of a chemical element, the magnetic quantum number refers to –

- (a) Orientation (b) Shape
(c) Size (d) Spin

I.A.S. (Pre) 2003

Ans. (a)

In atomic physics, the magnetic quantum number is the third set of quantum numbers (principal quantum number, azimuthal quantum number, magnetic quantum number and the spin quantum number) which describes the unique quantum state of an electron and its orientation. It is designated by the letter 'm'. The magnetic quantum number denotes the energy levels available within a subshell. This number yields the projection of the angular momentum corresponding to the orbital along a given axis. Principal quantum number (n) describes the electron shell or energy level of an electron. Azimuthal quantum number (l) describes the shape of a given orbital while spin quantum number (s) describes the spin of the electron within that orbital.

31. The number of electrons and neutrons in an element is 18 and 20 respectively. Its mass number is

- (a) 22 (b) 2
(c) 38 (d) 20
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (c)

Given that,
number of neutrons = 20
& number of electrons = 18
Thus, number of protons = number of electrons = 18
Mass number = number of protons + number of neutrons
= 18 + 20 = 38

32. What is the mass number of an element, the atom of which contains two protons, two neutrons and two electrons?

- (a) 2 (b) 4
(c) 6 (d) 8

43rd B.P.S.C. (Pre) 1999

Ans. (b)

The mass number of an element is the sum of a total number of protons and neutrons inside in its nucleus and represents by A. Therefore, mass number A = number of protons + number of neutrons. Therefore, mass number = 2 + 2 = 4

33. The mass number of a nucleus is :

- (a) the sum of the numbers of neutrons and protons
(b) the total mass of neutrons and protons
(c) always more than the atomic weight
(d) always less than its atomic number
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (a)

See the explanation of above question.

34. The number of neutrons in the nucleus of plutonium nuclide (${}_{94}\text{Pu}^{242}$) is :

- (a) 94 (b) 148
(c) 242 (d) 336
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (b)

The number of neutrons in the nucleus of plutonium nuclide (${}_{94}\text{Pu}^{242}$) is $242 - 94 = 148$. Because in ${}_{94}\text{Pu}^{242}$,
Number of protons = 94, and
Number of protons plus neutrons = 242

35. The atoms, in which the number of protons is same but the number of neutrons is different, are known as –

- (a) Isobars (b) Isomers
(c) Isotones (d) Isotopes

U.P.P.C.S. (Mains) 2009

Chhattisgarh P.C.S. (Pre) 2005

Uttarakhand U.D.A./L.D.A. (Pre) 2003

Ans. (d)

The Isotopes are a set of nuclides/atoms having the same number of protons, but a different number of neutrons. In other words, the same atomic number but having different atomic mass. Each individual isotope has a separate nuclei. The Isotopes that are unstable and undergo radioactive decay are called radioisotopes.

36. Isotopes of an element differ in :

- (a) the number of protons
(b) the mass number
(c) the number of electrons